



River Gauge

Final Report Document

Primary Contact

Roberta Hill

Apr 23, 2026

Project Team

Mihir Narayan- Project Lead

Will Jappe- ECE Technical Lead

Julia Faubus- Hardware Engineer

Luca Poulos- Hardware Engineer

Sachin Patel - ME Technical Lead

Isabella Valentino- Hardware Engineer

Shrey Abert- Hardware Engineer

Aram Najarian- Hardware Engineer

Nathaniel Xie- Software Engineer

Liliana Palomino- Hardware Engineer

Luca Pepe- Hardware Engineer

Abd AlRahman Abu Saleh - Hardware Engineer

Austin Fasnacht- Hardware Engineer

Liu von Engelbrechten - Hardware Engineer



TABLE OF CONTENTS

Goal	3
General Guidelines	3
Executive Summary	4
Purpose	4
Systems Overview	4
Mechanical Design	6
Assembly Overview	6
Electronics Enclosure	9
Critical Hardware Selection	15
Electrical Design	19
Design Overview	19
Circuit Board	20
Sensors	21
Appendix	24



Executive Summary

Purpose

River Gauge is a project focused on addressing the lack of accessible and affordable flood monitoring for smaller rivers, particularly in rural and underserved communities in Maine. Many existing river monitoring systems are expensive to install and maintain, which limits their deployment, while lower-cost alternatives often lack durability or real-time data access. As a result, communities are often underprepared for flash flood events due to limited flood awareness.

Generate supported this project by developing a low-cost, autonomous river monitoring system that can measure water levels, transmit data in real time to a publicly accessible website, and can survive in the harsh conditions seen in Maine over the winter. The team worked across mechanical and electrical subsystems to develop a solution that is both durable and easy to maintain. This was achieved by using a pressure sensor for water-level monitoring, developing a low-power LoRa communication system, and designing a modular frame, weatherproof enclosure, and anchoring system that can withstand harsh environmental conditions such as freeze-thaw cycles, debris impact, and wildlife interaction.

To achieve this, the team followed an iterative development process, beginning with subsystem research and early design, progressing through prototyping and integration, and working toward a minimum viable system capable of reliable data collection and transmission. The goal was to create a foundation for a scalable product that could be deployed across multiple sites with minimal installation effort.

Throughout the project, team members gained experience in interdisciplinary collaboration, system integration, and designing within real-world constraints, while developing technical and project management skills in a hands-on engineering environment.

Mihir Narayan
Project Lead
narayan.mih@northeastern.edu

Systems Overview

River Gauge is a low-cost, autonomous system designed to monitor water levels in small rivers and tributaries. The system uses a vented pressure sensor placed in the river to measure water level, with data transmitted wirelessly up to 2 km via LoRa. The electronics are designed for low-power operation, allowing the system to run for up to four months on two AA batteries, while continuously sending data to a publicly accessible website. The mechanical design separates the sensing and electronics components for durability and accessibility. An adjustable frame, similar to a tripod, is placed on the riverbank and supports a weatherproof electronics enclosure that allows easy battery maintenance. A tarp secured with carabiner clips provides additional protection from the elements. The sensor wire runs along the riverbed and is secured using rebar stakes, connecting to a custom sensor enclosure anchored into the riverbed with ground stakes. This setup allows for stable measurements while protecting the system from debris and environmental damage.



Figure 1: External View of River Gauge

The mechanical system provides a stable mounting structure and protective enclosures that allow the sensor and electronics to operate reliably in the river environment. The electrical system handles data collection from the pressure sensor, manages low-power operation, and



transmits data wirelessly to a remote server. The software processes this data and presents it on a public website, enabling users to access real-time river conditions.

Mechanical Design

Assembly Overview

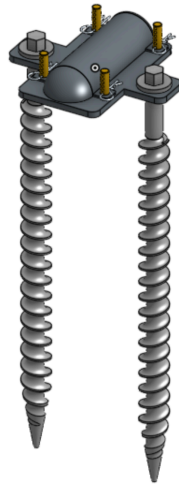
River Gaude is a low cost, highly accessible flood monitoring device that is designed to resist the harsh environments year around of rural Maine. The device is composed of 5 primary mechanical systems—stability framing, onshare electronics enclosure, river bed sensor enclosure, and both respective anchoring mechanisms.

During the assembly process the legs of the framing systems are unfolded and locked into the desired angle that aligns best with the ground. The duckbill anchors are then attached to the baseplate assembly using the quick links and pulled taught. The anchors are then driven into the ground to ensure stability. The electronics assembly is then bolted onto the baseplate, housing all electronic components as well as the battery compartment. From here the pressure sensor can be clipped into the protective enclosure and attached to riverbed baseplate using the cotter pins. The external wire is then run out of the electronics enclosure to the desired location in the river, preferably somewhere that won't be affected by debris. The penetrative ground anchors are then placed into the respective slots and screwed into the ground using the T-bar. The external wire is then anchored along the away using the wire anchors with the adjunct noncompressive clips. Once all this is complete the mast and snow tarp can be clipped onto the baseplate rounding out the installation process.

Maintenance of this system may involve the replacement of the batteries every few months, which involves unlatching the battery door with one quick release; located on the electronics enclosure. It also may require the electronics board to be accessed, which can be done by removing the large lid located on the side.



[Onshore Assembly]



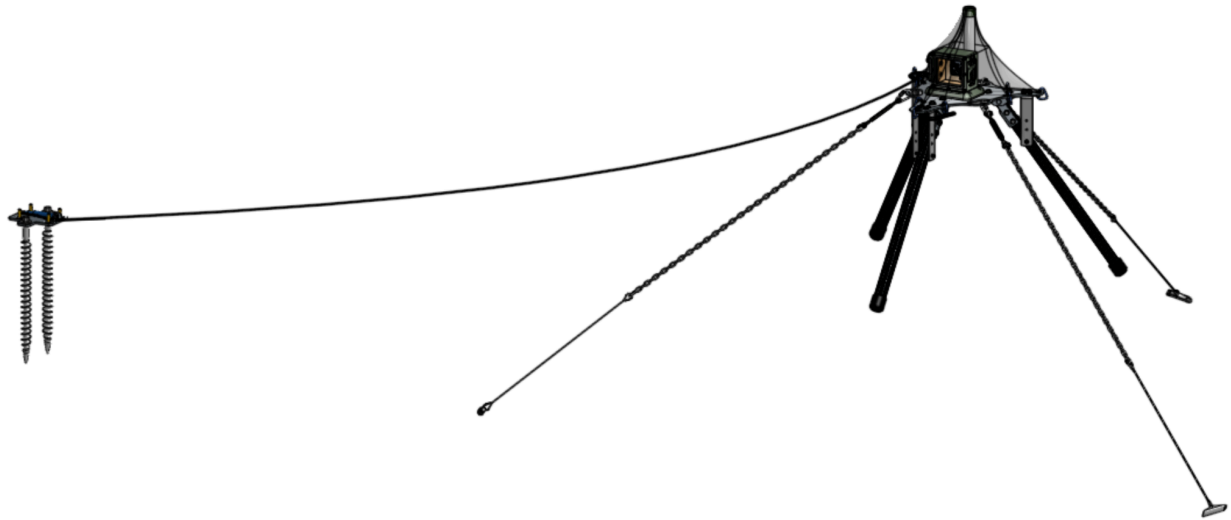
[Riverbed Sensor Assembly]

The system architecture emphasized modular field serviceability and installment through keeping tooling and hardware as similar as possible and cross function, a robust structure built to withstand the polarizing climate of Maine’s winters and summers, and a low cost manufacturable BOM to allow communities across the region to have better financial access to this product.

A large emphasis of our design was to minimize points in which water can pool. This meant minimizing open cavities, using appropriate IP ratings, and having robust sealing methods through gasket compression. We also utilized the “defense-in-depth” design philosophy which gave water flowing around our device direct run off paths and made it more difficult for water to seep in by creating multilayered non-uniform prints. Along with this we designed all parts that would be susceptible to thermal expansion with tolerance to allow for play while materials behave differently during different times of year.

Some potential shortcomings of this product include the assembled footprint. When assembled the foot print of the onshore assembly is very large. We chose this route to maximize the robustness of the apparatus during heavy storms or strong flooding. This large footprint does impede on the environment in a non-uniform geometry and is not harmfully invasive.

Another potential shortcoming of this product was the necessity of OTS parts that created tolerance stackups that had to be circumnavigated. In future iterations at a production scale more crucial components and connectors should be manufactured to ensure this reliability.



[Full Scale Assembly with external Wiring]



Electronics Enclosure

Components: Custom PETG FDM enclosure body, lid, and neoprene gasket; membrane vent, weatherproof cable glands, marine epoxy sealing system, spray paint topcoat, PETG filament, insulation, and stainless steel hardware

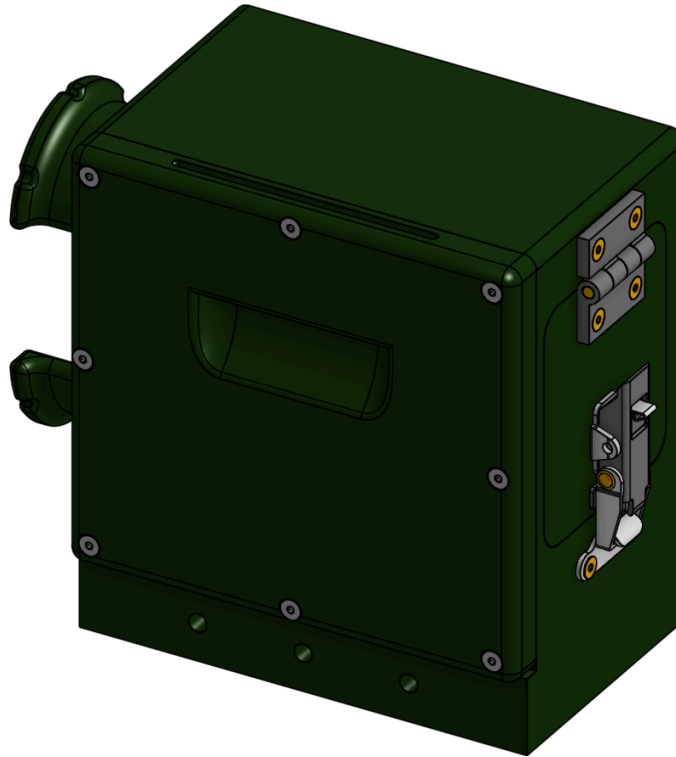
Off the Shelf Components:

- Hydrophobic membrane vent (pressure-equalizing)
- Weatherproof cable glands
- Neoprene sheet gasket material
- TotalBoat or equivalent marine-grade two-part epoxy
- Rust-Oleum or equivalent spray paint topcoat
- PETG filament
- Insulation material
- M4 brass heat-set inserts
- 316 Stainless Steel Hex-Drive Flat Head Screws
- Corrosion-Resistant 18-8 Stainless Steel Hex Nuts
- 18-8 Stainless Steel Hex Head Screws

Custom Components:

The primary function of the electronics enclosure is to house and protect the river gauge sensor electronics, including the LoRa 915 MHz radio module, battery, and associated circuitry during long-term outdoor deployment along Maine waterways. The enclosure must prevent water ingress during rain events and potential splash or temporary surface contact while remaining accessible for periodic field servicing. The design targets IP65 and NEMA 4X protection ratings, which require resistance to directed water jets and corrosive environmental exposure.

The enclosure was designed around a two-part body and lid configuration printed in PETG on a Bambu A1 FDM printer. PETG was selected over PLA for its superior moisture resistance, chemical resistance, and thermal stability under outdoor temperature cycling, and over ABS for its lower warping tendency and ease of FDM printing. An additional consideration was RF transparency, PETG does not significantly attenuate the 915 MHz LoRa signal, which was important given the antenna is housed internally.

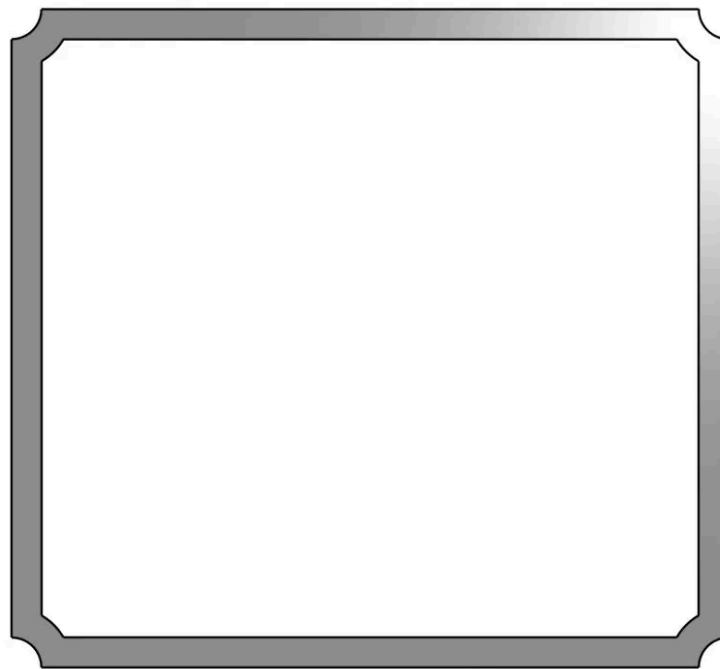


Full enclosure assembly

The lid-to-body interface uses a tongue-and-groove geometry to retain and compress a custom-cut neoprene sheet gasket. The tongue-and-groove profile was chosen over a flat-face gasket because it provides lateral constraint on the gasket, preventing blowout under pressure and improving sealing consistency across repeated assembly and disassembly cycles. Groove depth and tongue height were iterated to achieve approximately 20–25% gasket compression without requiring excessive lid closure force or causing the PETG features to deform. The gasket itself is cut from neoprene sheet stock to match the tongue-and-groove perimeter, making it a custom component despite sourcing from OTS sheet material. Lid retention is achieved through M4 brass heat-set inserts pressed into the enclosure body, providing durable metal threads that can withstand repeated fastener cycling without stripping the surrounding PETG.



Lid and body interface detail showing tongue-and-groove gasket channel



Gasket design

Print sealing is handled with marine epoxy applied across the exterior surfaces of the printed enclosure body and lid. FDM prints are inherently porous at a microscopic level, layer lines create small voids and pathways through the wall that liquid water can wick through over



time, even in a nominally watertight print. Applying epoxy over the outer PETG surfaces seals these layer lines, effectively eliminating the inherent porosity of the print and creating a continuous impermeable surface. The epoxy was applied in thin coats and sanded between layers to achieve a smooth, uniform finish with no air pockets or lifted edges.

A hydrophobic membrane vent is integrated into the enclosure to allow pressure equalization between the interior and external environment. Without venting, diurnal temperature cycling would create pressure differentials that could stress the lid seal or draw moisture inward through small imperfections in the sealing surfaces. The membrane passes air while blocking liquid water, maintaining the IP65 rating without internal pressure buildup. Cable penetrations are sealed using compression-style weatherproof cable glands, which provide a watertight seal around the wiring harness without requiring adhesives and allow cables to be removed and re-routed during servicing. Insulation material is included internally to protect electronics from temperature extremes during seasonal deployment.

All exposed fasteners, including the lid screws and any external mounting hardware are specified in 316 stainless steel or 18-8 stainless steel to resist corrosion in the wet, outdoor riverway environment. 316 stainless was selected for flat-head lid fasteners given its superior chloride resistance, while 18-8 hex head screws and nuts are used where direct water exposure is reduced.

The exterior of the enclosure is finished with a spray paint topcoat applied over the raw PETG surface, providing UV protection and a uniform appearance. The surface was lightly sanded before painting to improve adhesion. For a production iteration, a dedicated plastic-adhesion primer applied before the topcoat would improve long-term coating durability and reduce the risk of peeling under prolonged sun and moisture exposure.

Manufacturing Recommendations and Cost Estimates: PETG filament runs approximately \$20–\$30 per kilogram, with the enclosure body and lid together estimated to consume roughly 200–350 grams of filament depending on wall thickness and infill, placing material cost per unit at approximately \$5–\$12. Print time on the Bambu A1 is estimated at 6–10 hours total across both parts. Neoprene sheet gasket material is available through McMaster-Carr for approximately \$10–\$20 per sheet, sufficient for multiple gaskets. Marine epoxy runs approximately \$20–\$30 per cartridge with enough volume for many enclosures. M4 brass heat-set inserts are available in bulk for approximately \$5–\$10 per 50-count pack. Stainless steel fasteners run approximately \$5–\$15 depending on quantity and grade. Weatherproof cable glands run approximately \$1–\$3 each, and the membrane vent approximately \$3–\$8 each. Spray paint topcoat runs approximately \$8–\$12 per can. Total material cost per enclosure is estimated at \$40–\$70 depending on sourcing and quantity.



Electronics Stability Frame

- Component details (make one for each major component)
 - Image
 - Part Drawings may/may not be included
 - What is this design's function? Relate to the overall assembly
 - How did you get here in the design process?
 - Key features of design
 - Ex. Material Selection--why is this the best choice? Cost, performance, etc.
 - Ex. Tolerances
- Manufacturing recommendations
 - Include cost estimates if you can. Cite online resources or quotes from the manufacturer



Battery Removal

- Component details (make one for each major component)
 - Image
 - Part Drawings may/may not be included
 - What is this design's function? Relate to the overall assembly
 - How did you get here in the design process?
 - Key features of design
 - Ex. Material Selection--why is this the best choice? Cost, performance, etc.
 - Ex. Tolerances
 - Manufacturing recommendations
 - Include cost estimates if you can. Cite online resources or quotes from the manufacturer

External Wiring

- Component details (make one for each major component)
 - Image
 - Part Drawings may/may not be included
 - What is this design's function? Relate to the overall assembly
 - How did you get here in the design process?
 - Key features of design
 - Ex. Material Selection--why is this the best choice? Cost, performance, etc.
 - Ex. Tolerances
 - Manufacturing recommendations

Include cost estimates if you can. Cite online resources or quotes from the manufacturer



River Bed Sensor Anchoring

- Component details (make one for each major component)
 - Image
 - Part Drawings may/may not be included
 - What is this design's function? Relate to the overall assembly
 - How did you get here in the design process?
 - Key features of design
 - Ex. Material Selection--why is this the best choice? Cost, performance, etc.
 - Ex. Tolerances
- Manufacturing recommendations
 - Include cost estimates if you can. Cite online resources or quotes from the manufacturer

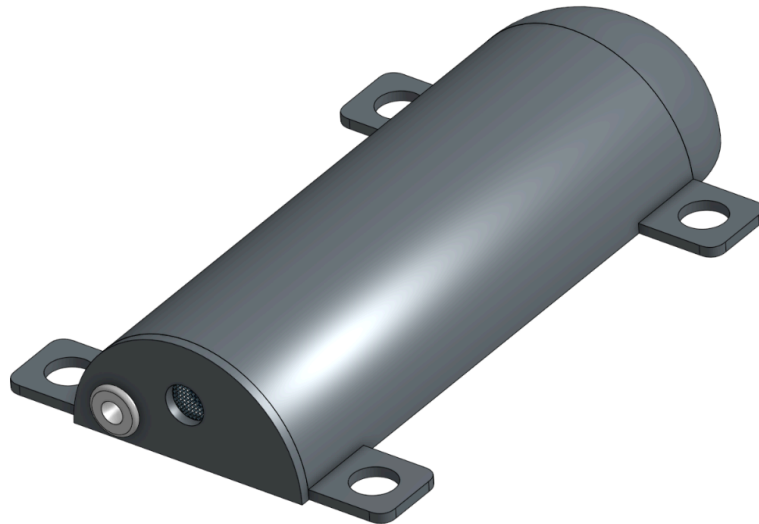
River Bed Sensor Enclosure — Component Details Report

Components: Steel enclosure tube, mesh end cap, mounting tabs, corrosion resistant coating system

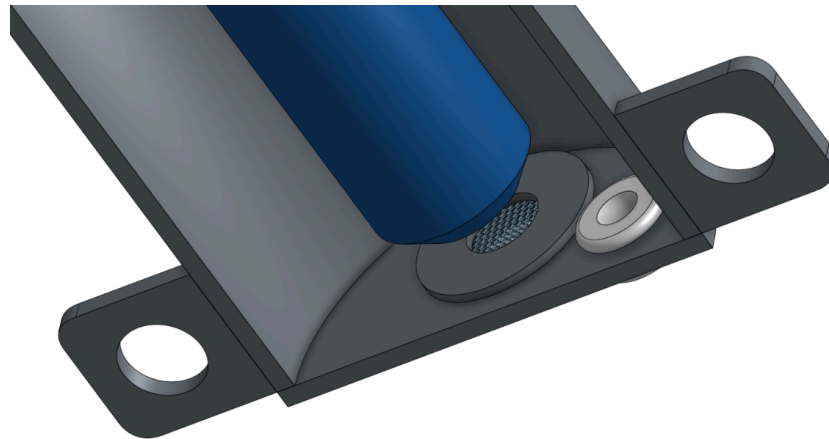
Off the Shelf Components:

- DOM mild steel tube — 2.75 in. internal diameter, 0.125 in. wall
- Stainless steel wire mesh
- Self-etching primer and epoxy spray paint
- Cable grommet and stainless steel sensor retention clip

Custom Components:



[Welded enclosure assembly]



[Mesh end cap]

The main function of the sensor enclosure is to house and protect a pressure sensor deployed permanently in a riverbed environment. The enclosure shields the sensor and its wiring from debris, water flow, and long-term corrosive wear while keeping the sensor accurately positioned in the water column. It connects to a permanent anchoring baseplate via welded mounting tabs and a clevis and cotter pin interface, allowing the enclosure to be lifted straight out of the water for servicing without disturbing the anchor.

The design started from a core serviceability requirement — the sensor needed to be removable in the field without digging up the riverbed anchor. This led to a sliding tube concept where the enclosure drops onto vertical guide rods and seats against a hard stop, retained by a cotter pin. Early concepts explored aluminum for its corrosion resistance, but aluminum TIG welding requires a level of operator training that was not available within the project schedule. The design was transitioned to DOM mild steel, which welds more easily and is structurally sound for this application. Corrosion resistance is handled by the coating system instead.

The enclosure tube is DOM mild steel, chosen because DOM tubing has a cleaner interior finish and more consistent wall thickness than standard ERW tube, which matters for fit and weld quality. The 2.75 inch internal diameter accommodates the pressure sensor and retention clip with approximately 0.010–0.020 inch of clearance, validated through full-scale 3D printed prototypes before any metal was cut. The cylindrical shape was intentional — it deflects debris more effectively than a rectangular cross section and avoids stress concentrations at corners.

The mesh end cap uses two layers of stainless steel wire mesh oriented at a 45 degree offset from each other. Stainless was selected specifically for the mesh because the high surface area of bare carbon steel mesh would corrode quickly in a river environment even with coating. The double-layer offset configuration was developed during fabrication — a single layer had too



much flex and larger openings than desired. Offsetting the two layers interlocks the geometry, effectively reducing aperture size and increasing stiffness without adding weight or a third layer. Welding thin mesh required minimum amperage tack welds rather than continuous beads to avoid burning through the wire.

The mounting tabs are small mild steel plates welded horizontally to the exterior of the tube. Their size made them prone to heat distortion during welding, which is something to account for in future fabrication. They interface with clevis and cotter pin hardware on the anchoring baseplate, providing a secure but removable connection.

After welding, the enclosure was deburred, sanded, and finished with a self-etching primer followed by rust-inhibiting epoxy spray paint. This two-step system bonds well to bare steel and provides adequate protection for freshwater submersion. For a production version, a two-part epoxy coating applied by brush would offer better long-term durability than rattle-can spray.

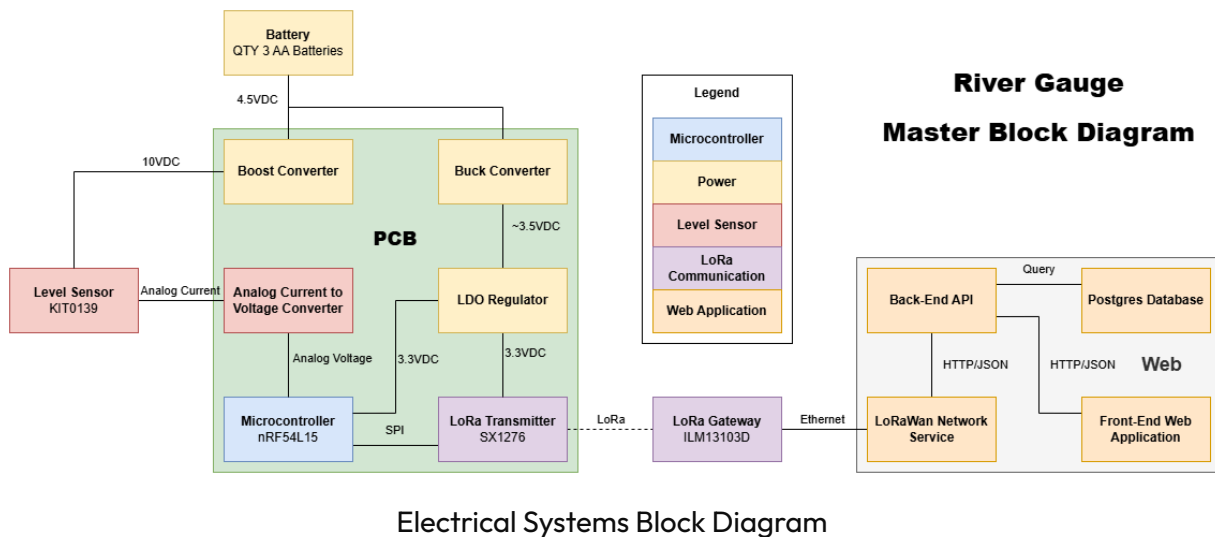
Manufacturing Recommendations and Cost Estimates:

DOM steel tubing (2.75 in. ID x 0.125 in. wall) is available through McMaster-Carr for approximately \$20–\$35 per foot depending on quantity. Stainless wire mesh is available through McMaster-Carr or Amazon for approximately \$10–\$20 per square foot. Self-etching primer runs about \$8–\$10 per can (Rust-Oleum, widely available). Epoxy topcoat runs approximately \$10–\$14 per can. TIG welding is the recommended process for all joints. A makerspace or community shop rate typically runs \$10–\$20 per hour for access. Total material cost for one enclosure is estimated at \$60–\$90 depending on sourcing.

Electrical Design

Design Overview

The primary electrical functions of the river gauge include the microcontroller, the power distribution, the level sensor, and the wireless transmitter. Once the data is received by a receiving device called a gateway, the data is then forwarded to a database to be viewed on the public web application.



Electrical Systems Block Diagram

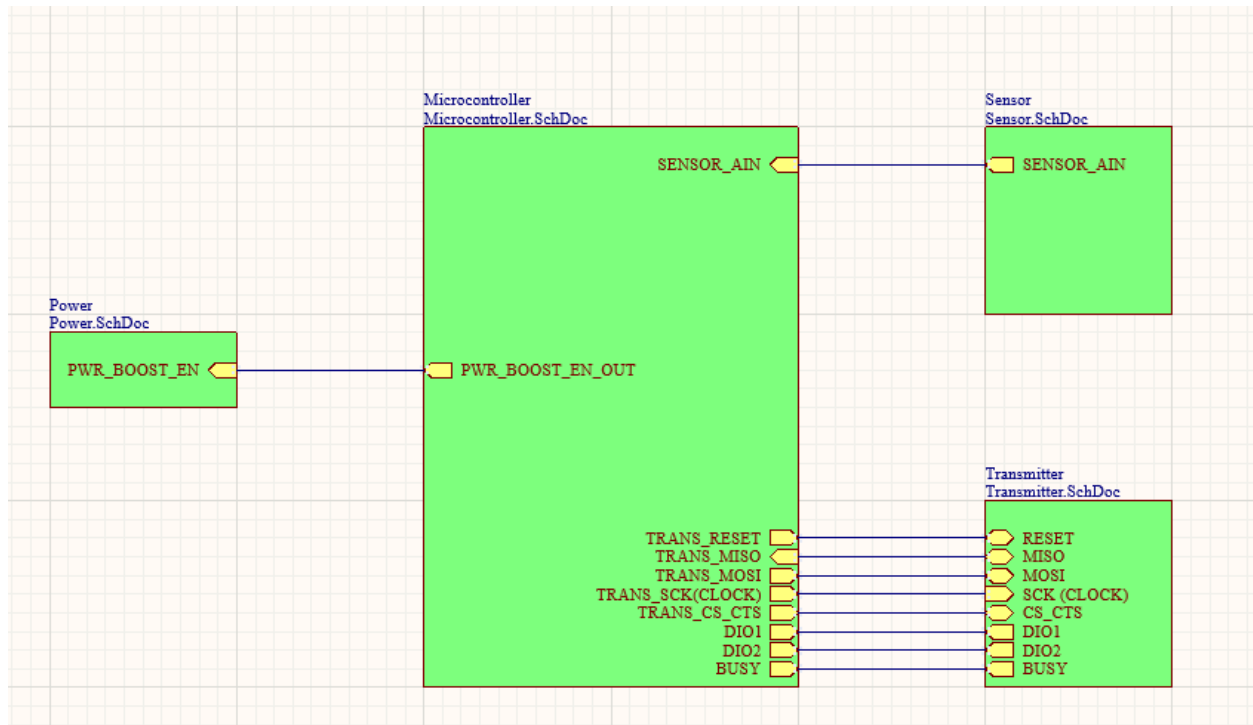
The PCB is securely attached inside of the electronics enclosure as described earlier, while the external battery pack is secured in a separate battery enclosure to be accessible when the batteries need to be replaced while still being safely secured when closed.

A Nordic microcontroller was utilized for this application due to the brand's reputation for low-power wireless applications. The microcontroller is put in a sleep state to significantly reduce power consumption. Three AA batteries power the board, which have to be replaced roughly 1 to 2 years. A waterproof pressure level sensor is used to measure the water level of the river, and is later transmitted by the LoRa transmitter.

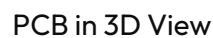
The gateway is intended to be installed inside a local building in the range of up to 4 miles from the gauge; that said, mileage (no pun intended) may vary depending on the terrain and weather conditions. Because the LoRa wireless standard is utilized, theoretical distances of hundreds of miles can be achieved under perfect conditions.

Circuit Board

The PCB was designed to house the microcontroller, the power distribution, the level sensor amplifier, and the LoRa transmitter. Connectors were included to interface with the level sensor, the battery pack, and an external antenna.



PCB Systems Overview



Sensors

- What is each sensor's purpose?
- Where should each sensor be placed?
- Include a link to the spec sheet and call out basic information that's relevant to the application like accuracy. Be sure to relate information to the application

21

Microcontroller Selection

The nRF54L15 was selected as the core microcontroller for the River Gauge system primarily due to its ultra-low power consumption. Since the device is deployed in a remote outdoor environment with no access to continuous power, minimizing energy draw is critical to extending battery life between maintenance cycles. The nRF54L15 supports multiple low-power sleep states, allowing the MCU to remain dormant between measurement intervals and only wake to sample the sensor, process data, and trigger a LoRa transmission. Key specs driving this decision include a supply voltage range of 1.7V–3.6V compatible with common field battery chemistries, a 128 MHz Arm Cortex-M33 core providing sufficient processing headroom for sensor math and LoRaWAN packet formatting, 256 KB RAM and 1 MB Flash adequate for the full software stack, and system-off current draw in the nanoamp range enabling months of operation on a single charge.

Subsystem Interfacing

The nRF54L15 interfaces with two external subsystems over SPI. The water pressure sensor is read on a scheduled wake interval — the MCU brings up the SPI peripheral, requests a reading, converts the raw pressure data into a water level value, then powers the bus back down. The LoRa radio module (SX1262) shares the SPI bus on a dedicated chip select line, with additional GPIO pins handling the BUSY line polled before issuing commands, a DIO1 interrupt line signaling transmission complete, and a RESET line. Once the pressure reading is packaged into a LoRaWAN payload, the MCU triggers a transmission, waits for the DIO1 interrupt confirming the packet was sent, then commands the radio back to sleep before returning itself to its low-power state.

For the future, it is better to put all of the SPI bus on the first channel instead of the second to avoid having to disable UART in order to use the SPI channel.

Power

The electrical system is powered by two AA alkaline batteries housed in a battery pack holder. The holder connects directly to the bottom right of the PCB board through a JST PH connector. The battery holder also includes an on and off switch that must be turned on to power the electrical system. The system is estimated to consume approximately 6684 mAh per year, while a typical AA alkaline battery provides 2500 mAh. Based on this, the batteries are expected to last 4 months under continuous use



before replacement is required. AA alkaline batteries are widely found at most stores such as Target and CVS.

The board consists of three voltage rails: 3.3 V, 4.5 V, and 12 V. The board takes in 4.5 V from the two batteries in series. A TPS62172DSGR buck converter steps this voltage down to 3.3 V to power the main circuitry including the chip. A TLV61047DDCR boost converter steps the 4.5 V input up to 12 V to power the pressure sensor. A reverse polarity protection circuit is included to prevent damage if the alkaline batteries are placed incorrectly in the holder.

The power system incorporates both a temperature and voltage monitoring system. The temperature sensor is monitored using the TMP303ADRLR, which asserts a fault when the temperature exceeds its fixed threshold of approximately 105 degrees Celsius. It will then return to operating near 95 degrees Celsius. This ensures that it will only turn on during safe conditions.

The voltage monitoring is implemented using the TPS3840PL30DBVR. This device has a nominal threshold of 3.0 V with up to $\pm 1.5\%$ accuracy. When it is used to monitor a 4.5 V voltage rail, this corresponds to a trip range of approximately 4.43 V to 4.57 V. The system is therefore turned off when the batteries are close to dead.

The temperature sensor outputs a low signal during normal (safe) temperature conditions, and a high signal when an overtemperature fault is detected. This signal is then passed through an inverter so that a high signal represents a safe temperature. This inverted signal is connected to the manual reset (MR) pin of the voltage monitor which is active low. As a result, the temperature fault drives the MR pin low and forces a reset, disabling the system. When the temperature is within a safe range, the MR pin remains high, allowing normal operation.

Appendix

